

Naman Goyal

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EDUCATION

Indian Institute of Technology Delhi

CGPA: 9.22/10 (Dean's List)

B.Tech. in Electrical Engineering | **Department Rank: 1**

Jul. 2024 – May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Digital Electronics, Probability.

HONORS & AWARDS

- **Informatics Olympiads:** USA Computing Olympiad (USACO) **Gold; Rank 1** in Croatian Open in Informatics.
- **International Physics Olympiad (IPhO): Gold Medalist** at INPhO and ranked **top 37** nationally.
- **International Astronomy Olympiad (IOAA): Gold Medalist** at INAO and ranked **top 29** nationally.
- **International Mathematics Olympiad (IMO): Gold Medalist** at INMO and invited to the IMO Training Camp.
- **Panini Linguistics Olympiad (PLO): Bronze Medalist** in the second round of PLO.

EXPERIENCE

CISPA Helmholtz Center for Information Security

Saarbrücken, Germany

Research Intern, Trusted Systems Group

May 2026 – Jul. 2026

- Audited 131 open-source RISC-V cores, grading RTL, testbenches, and commits against security claims.
- Built a reproducible Python pipeline yielding **50+ figures** and two composite hardware-security indices.
- Found **95%** of cores lacked side-channel defenses and **77%** lacked secure boot and co-authoring a publication.

PROJECTS

FPGA-Accelerated Tick-to-Trade Engine

C++20, SystemVerilog, FPGA, DMA

- Parsed Ethernet/IP/UDP and market-data frames in FPGA fabric, with no CPU in the critical path.
- Ingested FPGA output over **DMA** via a busy-pollled **lock-free SPSC ring buffer**, avoiding syscalls.
- Reached deterministic **sub-microsecond** tick-to-decision latency by keeping hot logic in hardware.

Low-Latency Limit Order Book & ITCH Feed Handler

Modern C++20, Linux

- Decoded **NASDAQ ITCH 5.0** over UDP multicast (MoldUDP64) across all **23 message types**.
- Built side-separated price-level trees with FIFO queues and an **O(1)** hashmap for cancels/modifies.
- Engineered a **zero-allocation** hot path (slab pools); sustained **10M+ msgs/s** at ~100 ns median latency.

Wire-Speed FPGA Network Parser & Hardware Trigger

SystemVerilog, Vivado

- Built a pipelined parser validating MAC/IP/UDP headers, extracting fields at **one beat per clock**.
- Added a **fixed-latency Level-1 trigger** making deterministic accept/reject decisions inline.
- Verified with a self-checking testbench; characterized timing closure and **LUT/FF/BRAM** usage.

KAN-PINN Battery State-of-Health Model

PyTorch, Physics-Informed ML

- Built a **Kolmogorov-Arnold + physics-informed** network for battery State-of-Health prediction.
- Enforced degradation physics via an **augmented-Lagrangian** loss; validated on MIT datasets.

QicQacQoe — Quantum Game Simulator

Qiskit, Quantum Computing

- Built a **13-qubit** state-vector simulator (Hadamard/CNOT) over superposed, entangled board states.
- Added measurement and **state-collapse** logic to score win conditions across the entangled space.

TECHNICAL SKILLS

- **Languages and Libraries:** C++, SystemVerilog/Verilog, Python, JavaScript, MATLAB, PyTorch, Qiskit, NumPy
- **Tools:** Vivado, CMake, Git, GDB, perf, Linux, Docker